ECON 0100 | Exercise C3 | Corrective Policies

Solutions

Answers and work are shown in the red blocks.

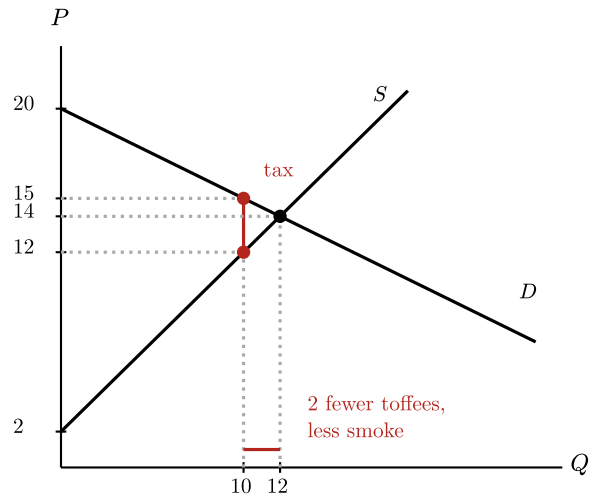
Q1 | The Effects on Residents

Smoked toffee again, with the (same curves and 3 Galleon externality as Exercise C2).

$$D : P_b = 20 - \frac{1}{2}Q_d \qquad S : P_s = 2 + Q_s$$

a) This policy had an unintended consequence. The proliferation of smoked toffee makers meant the town saw a subsequent rise in *smoke*. Wood smoke is harmless at low levels, but can cause lung issues at the higher concentrations experienced in Hogsmeade. Use a graph to discuss the unintended effect the tax had on air quality. No need to discuss welfare.

a) The tax cut the quantity of smoked toffee from 12 to 10, so less wood smoke: air quality improved.



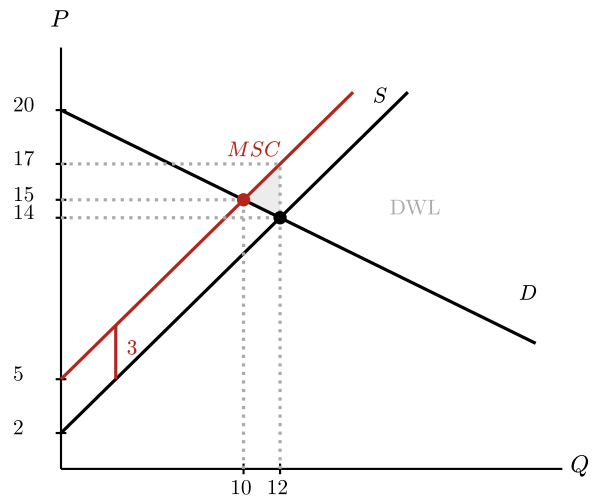
Smoke comes from *making* toffee, so the amount of smoke moves with the quantity produced. Exercise C2 found the tax equilibrium: without the tax $20 - \frac{1}{2}Q = 2 + Q$ gives $Q = 12$; with the 3-Galleon wedge,

$$20 - \tfrac{1}{2}Q = (2 + Q) + 3 \quad \Rightarrow \quad Q = 10$$

Two fewer toffees are made, so less wood is burned and less smoke reaches the town. The Ministry taxed toffee for wizards' metabolic health, but as a side effect it also cleaned up the air, an unintended consequence that helps residents.

b) Public health officials have estimated the externality to be equal to 3 Galleons per unit of smoked toffee produced. Use a graph to find the socially efficient quantity of smoked toffee and deadweight loss associated with the wood smoke.

b) Socially efficient quantity: 10. DWL: 3 Galleons.



The smoke is a cost the sellers do not pay, so the cost to society of each toffee is the seller's marginal cost plus 3:

$$MSC = (2 + Q) + 3 = 5 + Q$$

The socially efficient quantity is where marginal benefit meets marginal social cost:

$$20 - \tfrac{1}{2}Q = 5 + Q \quad \Rightarrow \quad Q = 10, \quad P = 15$$

Left alone, the market makes 12 toffees. The 11th and 12th toffees are worth less to buyers than they cost society: at $Q = 12$ a buyer values the last toffee at 14, but it costs 17 to make once the smoke is counted. The deadweight loss is the triangle between MSC and D over those two toffees:

$$\text{DWL} = \tfrac{1}{2} \times (17 - 14) \times (12 - 10) = 3$$

The efficient quantity is exactly the tax equilibrium quantity from Exercise C2, and the buyer price there, 15, is the price here. That is no coincidence: the tax is 3 and the smoke costs 3. With the tax, a seller needs the buyer's price to cover their own cost *plus* 3, which is the same as covering the full social cost of the toffee. The tax makes sellers act as if they paid for their own smoke, so the market stops at the efficient quantity and the deadweight loss is gone.

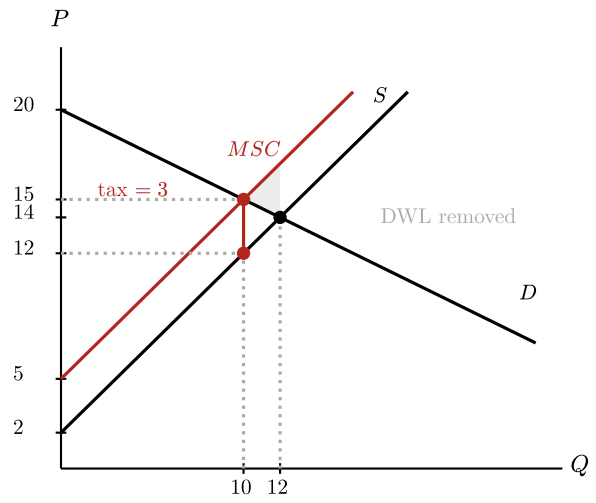
Q2 | Policy Proposal

What type and size of policy would you propose to eliminate the DWL you identified in Q1?

Policy Type: _____ Policy Size: _____

Buyer Price: _____ Seller Price: _____ Post-Policy Quantity: _____

Policy Type: a tax on smoked toffee. Policy Size: 3 Galleons per toffee. Buyer Price: 15. Seller Price: 12. Post-Policy Quantity: 10.



From Q1, the market makes too much toffee: $20 - \frac{1}{2}Q = 2 + Q$ gives $Q = 12$ at $P = 14$, while the socially efficient quantity, where D meets $MSC = 5 + Q$, is $Q = 10$. The deadweight loss is $\frac{1}{2} \times 3 \times (12 - 10) = 3$.

A negative externality calls for a tax equal to the external cost, 3 per toffee. Then the buyer price sits 3 above the seller price:

$$P_b = P_s + 3 \quad \Rightarrow \quad 20 - \tfrac{1}{2}Q = (2 + Q) + 3 \quad \Rightarrow \quad Q = 10, \quad P_b = 15, \quad P_s = 12$$

The market now stops at 10, the efficient quantity, so the deadweight loss triangle is gone. The seller keeps 12, their own marginal cost of the 10th toffee; the buyer pays 15, its marginal social cost. Each toffee is made only if a buyer values it at least as much as it costs society, smoke included.

Q3 | Deadweight Loss Intuition

If the marginal cost of a gallon of gas is \$4.00, the negative externality is \$1.00, and the marginal benefit is \$4.50, what is the deadweight loss?

Deadweight Loss: _____

> Deadweight Loss: \$0.50.
>
> The buyer values the gallon at \$4.50 and it costs the seller \$4.00, so the two of them gain \$0.50 and the gallon is bought. But the gallon also costs everyone else \$1.00 in pollution, so its marginal social cost is $4.00 + 1.00 = 5.00$, more than the \$4.50 it is worth. Counting everyone, this gallon makes society $4.50 - 5.00 = -0.50$ worse off. That \$0.50 is the deadweight loss: the surplus society would gain if this gallon were not sold.